



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201

ENGINEERING MATHEMATICS II

UNIT 3 –VECTOR DIFFERENTIAL CALCULUS

LP11- DIRECTIONAL DERIVATIVE -DIVERGENCE OF A

VECTOR FIELD-CURL OF A VECTOR FIELD

Directional Derivative (D.D) of a scalar point function

The derivative of a point function (scalar or vector) in a particular direction is called its directional derivative along the direction.

The directional derivative of a scalar function φ in a given direction $\vec{a}$ is the rate of change of φ in that direction. It is given by the component of $\nabla\varphi$ in the direction of $\vec{a}$.

The directional derivative of a scalar point function in the direction of $\vec{a}$ is given by

$$\text{D.D} = \frac{\nabla\varphi \cdot \vec{a}}{|\vec{a}|}$$

The maximum directional derivative is $|\nabla\varphi|$ or $|\text{grad } \varphi|$.

Problems based on Directional Derivative

Example: 2.6 Find the directional derivative of $\varphi = 4xz^2 + x^2yz$ at $(1, -2, 1)$ in the direction of $2\vec{i} - \vec{j} - 2\vec{k}$.

Solution:

$$\text{Given } \varphi = 4xz^2 + x^2yz$$

$$\nabla\varphi = \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z}$$

$$= \vec{i}(2xyz + 4z^2) + \vec{j}(x^2z) + \vec{k}(x^2y + 8xz)$$

$$\therefore (\nabla\varphi)_{(1,-2,-1)} = 8\vec{i} - \vec{j} - 10\vec{k}$$

$$\text{Given } \vec{a} = 2\vec{i} - \vec{j} - 2\vec{k}$$

$$|\vec{a}| = \sqrt{4 + 1 + 4} = 3$$

$$\text{D. D} = \frac{\nabla\varphi \cdot \vec{a}}{|\vec{a}|}$$

$$= (8\vec{i} - \vec{j} - 10\vec{k}) \cdot \frac{(2\vec{i} - \vec{j} - 2\vec{k})}{3}$$

$$= \frac{1}{3} (16 + 1 + 20) = \frac{37}{3}$$

Example: 2.10 Find the directional derivative of $\varphi = xy + yz + zx$ at $(1, 2, 0)$ in the direction of $\vec{i} + 2\vec{j} + 2\vec{k}$. Find also its maximum value.

Solution:

$$\text{Given } \varphi = xy + yz + zx$$

$$\nabla\varphi = \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z}$$

$$= \vec{i}(y + z) + \vec{j}(x + z) + \vec{k}(y + x)$$

$$\therefore (\nabla\varphi)_{(1, 2, 0)} = 2\vec{i} + \vec{j} + 3\vec{k}$$

$$\text{Given } \vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$$

$$|\vec{a}| = \sqrt{1 + 4 + 4} = 3$$

$$\text{D. D} = \frac{\nabla\varphi \cdot \vec{a}}{|\vec{a}|}$$

$$= \frac{(2\vec{i} + \vec{j} + 3\vec{k}) \cdot (\vec{i} + 2\vec{j} + 2\vec{k})}{3}$$

$$= \frac{1}{3} (2 + 2 + 6) = \frac{10}{3}$$

Maximum value is $|\nabla\phi| = \sqrt{4 + 1 + 9} = \sqrt{14}$

DIVERGENCE, CURL – IRROTATIONAL AND SOLENOIDAL VECTORS

Divergence of a vector function

If $\vec{F}(x, y, z)$ is a continuously differentiable vector point function in a given region of space, then the divergence of $\vec{F}$ is defined by

$$\begin{aligned}\nabla \cdot \vec{F} = \text{div } \vec{F} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}) \\ \text{div } \vec{F} &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \text{ where } \vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}\end{aligned}$$

Note: $\nabla \cdot \vec{F}$ Is a scalar point function.

Solenoidal vector

A vector $\vec{F}$ is said to be solenoidal if $\text{div } \vec{F} = 0$ (i.e) $\nabla \cdot \vec{F} = 0$

Curl of a vector function

If $\vec{F}(x, y, z)$ is a differentiable vector point function defines at each point (x, y, z) in some region of space, then the curl of $\vec{F}$ is defined by

$$\begin{aligned}\text{Curl } \vec{F} = \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \vec{i} \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) - \vec{j} \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z} \right) + \vec{k} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)\end{aligned}$$

Where $\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}$

Note: $\nabla \times \vec{F}$ Is a vector point function.

Irrotational vector

A vector is said to be irrotational if $\text{Curl } \vec{F} = 0$ (i.e) $\nabla \times \vec{F} = 0$

Example: 2.21 If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ then find $\text{div } \vec{r}$ and $\text{curl } \vec{r}$

Solution:

$$\text{Given } \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\text{Now } \text{div } \vec{r} = \nabla \cdot \vec{r}$$

$$= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1 + 1 + 1 = 3$$

$$\text{And } \text{curl } \vec{r} = \nabla \times \vec{r}$$

$$\nabla \times \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

$$= \vec{i} \left(\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(y) \right) - \vec{j} \left(\frac{\partial}{\partial x}(z) - \frac{\partial}{\partial z}(x) \right) + \vec{k} \left(\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) \right)$$

$$= \vec{i}(0) + \vec{j}(0) + \vec{k}(0) = \vec{0}.$$

Example: 2.22 If $\vec{F} = xy^2\vec{i} + 2x^2yz\vec{j} - 3yz^2\vec{k}$ find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$ at the point (1,-1, 1).

Solution:

$$\text{Given } \vec{F} = xy^2\vec{i} + 2x^2yz\vec{j} - 3yz^2\vec{k}$$

$$\begin{aligned} \text{(i) } \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(2x^2yz) + \frac{\partial}{\partial z}(-3yz^2) \\ &= y^2 + 2x^2z - 6yz \end{aligned}$$

$$\nabla \cdot \vec{F}_{(1,-1,1)} = 1 + 2 + 6 = 9$$

$$\begin{aligned} \text{(ii) } \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & 2x^2yz & 3yz^2 \end{vmatrix} \\ &= \vec{i} \left[\frac{\partial(-3yz^2)}{\partial y} - \frac{\partial(2x^2yz)}{\partial z} \right] - \vec{j} \left[\frac{\partial(-3yz^2)}{\partial x} - \frac{\partial(xy^2)}{\partial z} \right] + \vec{k} \left[\frac{\partial(2x^2yz)}{\partial x} - \frac{\partial(xy^2)}{\partial y} \right] \\ &= \vec{i}(-3z^2 - 2x^2y) - \vec{j}(0) + \vec{k}(4xyz - 2xy) \end{aligned}$$

$$\begin{aligned} \nabla \times \vec{F}_{(1,-1,1)} &= \vec{i}(-3 + 2) + \vec{k}(-4 + 2) \\ &= -\vec{i} - 2\vec{k} \end{aligned}$$

Example: 2.34 If $\vec{F} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + \lambda z)\vec{k}$ is solenoidal, then find the value of λ .

Solution:

Given $\vec{F}$ is solenoidal.

$$(ie) \nabla \cdot \vec{F} = 0$$

$$\Rightarrow \frac{\partial}{\partial x}(x + 3y) + \frac{\partial}{\partial y}(y - 2z) + \frac{\partial}{\partial z}(x + \lambda z) = 0$$

$$\Rightarrow 1 + 1 + \lambda = 0$$

$$\therefore \lambda = -2$$

Example: 2.32 Prove that the vector $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ is solenoidal.

Solution:

$$\text{Given } \vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$$

To prove $\nabla \cdot \vec{F} = 0$

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(z) + \frac{\partial}{\partial y}(x) + \frac{\partial}{\partial z}(y) \\ &= 0 \end{aligned}$$

Example: 2.36 Show that the vector $\vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational.

Solution:

$$\text{Given } \vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$$

To prove $\text{curl } \vec{F} = 0$

(i.e) To prove $\nabla \times \vec{F} = 0$

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + z^3 & 3x^2 - z & 3xz^2 - y \end{vmatrix} \\ &= \vec{i}(-1 + 1) - \vec{j}(3z^2 - 3z^2) + \vec{k}(6x - 6x) = \vec{0}\end{aligned}$$

$\therefore \vec{F}$ is irrotational.

Example: 2.37 Find the constants a, b, c so that the vectors is irrotational

$$\vec{F} = (x + 2y + az)\vec{i} + (bx + 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}.$$

Solution:

Given $\vec{F} = (x + 2y + az)\vec{i} + (bx + 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational.

(ie) $\nabla \times \vec{F} = 0$

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx + 3y - z & 4x + cy + 2z \end{vmatrix} = \vec{0} \\ \Rightarrow \vec{i}(c + 1) - \vec{j}(4 - a) + \vec{k}(b - 2) &= \vec{0} \\ \Rightarrow c + 1 = 0 ; \quad 4 - a = 0 ; \quad b - 2 = 0 \\ \Rightarrow c = -1 ; \quad 4 = a ; \quad b = 2\end{aligned}$$

Exercise:

1. A skier is on a mountain with equation $z = 100 - 0.4x^2 - 0.3y^2$ where z denotes height.

Questions:

- a) The skier is located at the point with xy -coordinates $(1, 1)$, and wants to skier downhill along the steepest possible path. In which direction (indicated by a vector (a, b) in the xy -plane) should the skier begin skiing?
 - b) The skier begins skiing in the direction given by the xy -vector (a, b) you found in part (a), so the skier heads in a direction in space given by the vector (a, b, c) . Find the value of c .
2. Consider a fluid flowing through a pipe with a cross-sectional area those changes along the length of the pipe. The velocity of the fluid at any point in the pipe can be represented by a vector function $F(x, y, z) = \langle u(x, y, z), v(x, y, z), w(x, y, z) \rangle$, where u , v , and w are the components of the velocity vector in the x , y , and z directions, respectively.

Question:

Find the divergence of the vector function $F(x, y, z) = \langle x^2, y^3, z^4 \rangle$ at the point $(1, 2, 3)$.

3. The hydraulic actuator system is able to generate precise forces and torques to move heavy loads and control the machine's movements. It is controlled by a computer-controlled servo valve. The servo valve is controlled by a computer system that sends precise signals to the valve, which adjusts the flow of hydraulic fluid into and out of the cylinder to precisely control the position and force of the piston. A vector $A = xz \mathbf{i} - 2yz \mathbf{j} + 2xy \mathbf{k}$ to represent the force generated by the hydraulic cylinder at a given position and fluid pressure, where x , y , and z represent the position of the piston.

Question:

Calculate curl of the force generated by the hydraulic cylinder.

4. Suppose we have a fluid flow in a region of space that is described by the velocity vector field: $v(x, y, z) = (x + 2y + 4z) \mathbf{i} + (2ax + by - z) \mathbf{j} + (4x - y + 2z) \mathbf{k}$ where x , y , and z are the Cartesian coordinates in the region. This vector field represents a fluid flow that varies with position, with faster flow near the center of the region and slower flow near the edges.

Question:

Find the value of a and b if the vector v is both solenoidal and irrotational.

Discussion Questions

Try to answer the following questions

1. What is the application of directional derivative?
2. Is the curl of a constant vector is zero?
3. What is the physical interpretation of a divergence of a vector?